

INDIVIDUAL RESEARCH PLAN
OF THE PhD STUDENT OF THE FIRST DOCTORAL SCHOOL
OF THE MEDICAL UNIVERSITY OF GDAŃSK

professional title, given name and surname of the PhD student:	M.Sc. Jakub Pawłowski
current year of studies:	I year
supervisor(s):	Dr hab. Anna Supernat
assistant supervisor:	Dr Krzysztof Pietruczuk
MUG research unit of the PhD student:	Division of Translational Oncology
title of the doctoral dissertation:	PL: "Opracowanie i walidacja multimodalnego modelu uczenia maszynowego do diagnozowania depresji"
	EN: "Development and Validation of a Multimodal Machine Learning Model for Diagnosing Depression"
Language of dissertation preparation:	Polish ☐ English ☐

I. TIME SCHEDULE OF PREPARING THE DOCTORAL DISSERTATION:

Academic year	Semester	Description
2024/2025 1st year: progress on the dissertation made in the 1st year	1 st -2 nd	Conducted an in-depth literature review on multimodal ML for depression diagnosis and related clinical endpoints. Tested key technologies for data collection and processing (audio capture, transcription, secure storage), and mapped ethics/RODO requirements (including consent design and draft DPIA). Consulted with experts in AI and psychiatry; defined research questions, hypotheses, and preliminary evaluation metrics. Designed the architecture and user flows of a crowdsourcing platform for audio data collection; prepared the initial project documentation.
2025/2026	3 rd	Develop and implement the crowdsourcing platform for gathering audio data, integrate transcription tools (e.g., models like Whisper for audio-to-text conversion), and perform initial data preprocessing, quality assurance, and ethical compliance checks (e.g., RODO). Begin systematic data collection; establish versioned data schemas and metadata standards.
	4 th	Analyze collected datasets, develop a prototype of the multimodal model (focusing on audio features combined with text

		transcriptions, incorporating CNN architectures for audiogram analysis and text-based NLP models), and conduct preliminary testing, debugging, and performance evaluations.
2026/2027	5 th	Expand the training dataset; perform iterative model validation and fine-tuning (fusion strategies, regularization, calibration). Solidify the statistical analysis plan (primary/secondary endpoints, CI/uncertainty, robustness checks). Prepare manuscript drafts to be subsequently published in peer-reviewed journals.
	6 th	Initiate clinical validation with collaborating medical sites; harmonize protocols and data capture forms. Formally state and test the research hypotheses defined at the outset; document methodology and analytical workflow reproducibly. Begin writing the dissertation monograph with the Introduction and Methods chapters; update ethics/RODO documentation as needed; refine the model based on clinical feedback.
2027/2028	7 th	Finalize model integration; conduct comprehensive robustness studies (cross-dataset evaluations, subgroup analyses, sensitivity/stress tests). Update and revise manuscript drafts (and submit/resubmit where feasible) to support dissemination; manuscripts are supportive to the monograph, which remains the primary dissertation form. Compile the full dissertation draft (monograph), integrating results, discussion, limitations, and future work.
	8 th	Revise and polish the dissertation based on supervisors' feedback; finalize figures, appendices, and references. Prepare defense materials (abstracts, presentation, lay summary); complete administrative steps and submit the final dissertation for review.

II. END DATE FOR SUBMISSION OF THE DOCTORAL DISSERTATION:

May 30, 2028

III. THE LEARNING OUTCOMES ATTAINED THROUGH WORK ON THE PREPARATION OF THE DOCTORAL DISSERTATION – in accordance with appendix No. 1.

IV. IS IT PLANNED TO APPLY FOR RESEARCH GRANTS:

☐ YES If 'Yes': by when:
 grant type:
☐ NO

.....
 full hand-written name of the PhD student

V. DECLARATION OF THE SUPERVISOR(S):

I here declare that the content of points I-III of the Individual Research Plan has been agreed with me.

.....
Supervisor's signature and stamp

.....
Second supervisor's signature and stamp¹⁾

VI. OPINION OF THE ASSISTANT SUPERVISOR²⁾:

The proposed Individual Research Plan is well-structured and realistic, aligning with the objectives of developing a multimodal machine learning model for depression diagnosis through crowdsourced audio and text data. The emphasis on platform development, data integration with CNN for audiograms, and clinical validation is appropriate. I recommend close monitoring of data collection ethics (e.g., RODO compliance) and iterative model testing to ensure robustness. This plan supports timely completion within the doctoral program.

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Assistant supervisor's signature and stamp²⁾

VII. ENCLOSURES:

1. The learning outcomes attained in the work on the preparation of the doctoral dissertation³⁾
2. Other:

1) In the event the PhD student has a second supervisor

2) In the event the PhD student has an assistant supervisor

3) In appendix No. 1, tick the learning outcomes attained in working on the preparation of the doctoral dissertation.